Low-cost air pollution monitoring systems

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***Abstract*— Recently, discussions on air pollution have escalated due to its harmful impact on human health. More and more often we meet a remarkable interest in air quality monitoring, the society we are living in is increasingly approaching methods of acquiring and installing weather and air quality monitoring stations in different locations, whether we are talking about urban or rural areas. While traditional air quality monitoring systems are slightly outdated and they have limitations in providing real-time data at a reasonable cost, technological advances such as the Internet of Things (IoT) offer promising solutions for efficient monitoring in terms of costs, time and energy consumption. Thus, currently smart cities manage to successfully integrate both hardware and software devices to establish complete air quality monitoring systems. This paper proposes an extensive state-of-the-art and a synthesis on the environmental parameters and the air quality index obtained through certain current methods of air quality monitoring.**

To overcome today's pollution monitoring systems drawbacks, the air quality index (AQI) was defined according to EU and World Health Organization standards. The AQI was computed based on three different databases in a proposed online application for pollution monitoring and analysis. The calculation method of AQI with existing monitoring systems was compared with current standards AQI for a clear and uniform view of air quality.

Keywords—air pollution, air quality index monitoring, low cost monitoring system

I. Introduction (*Heading 1*)

The expansion of urban populations, fueled by factors like enhanced quality of life and citizen migration, has transformed cities into expansive megacities. As a result, various environmental challenges have emerged that urban environments are struggling to solve, failing to meet all the demands of contemporary society. Numerous studies and research results suggest that by 2050, 75% of the world's population will have moved to the urban environment, and this accelerated urbanization is expected to have negative effects on the environment [1].

Recognizing this reality, the adoption of a science and technology-driven approach emerges as the cornerstone for the advancement of smart cities [2].

The increase in air pollution, attributed to urban agglomeration, rapid industrialization, and the expansion of transportation networks, has made both public and governmental entities pay more attention to this topic. A widely recognized consequence of prolonged exposure to air pollution is the occurrence of serious respiratory diseases [3].

In this context, smart cities represent a paradigm shift in urban planning, using cutting-edge technologies to address the multiple challenges associated with burgeoning urbanization. By incorporating advanced solutions such as Internet of Things (IoT) devices, data collection and analysis, and artificial intelligence, smart cities aim to increase efficiency, resource management, air quality, and the overall quality of life for their residents.

The current problem is that although every day more and more people (whether we mean researchers or citizens) are turning their attention to this topic of air quality and more and more data sources are appearing on the Internet from various smart devices or air quality monitoring, however, these systems, without very specific documentation from those providing the data, cannot guarantee the accuracy of the information, the way in which the data were collected, nor the method of calculating the air quality index.

Thus, the topic of interest in this paper is related to how the indices are calculated and whether the measurements and conclusions are according to specific standards. For this study we used 3 sources of air quality data, which we will talk about in detail later in this paper.

This paper is structured in five sections, starting with the context of this study presented in introduction, followed by a state of the art in monitoring air quality, in section 2, and the computation accepted methods for expressing the air quality index according to different standards in section 3. The design and features of the developed application for AQI monitoring is presented in section 4, followed by a conclusion section.

II. MONITORING AIR QUALITY

Air pollution is an important concern, especially in urban area, due to its significant impact on the health of the entire community. The partial access to unvalidated data, the high cost and the non-scalability of the conventional air monitoring system force researchers to try to develop and innovate in air pollution monitoring systems. Advanced technologies such as Internet of Things (IoT), Wireless Sensor Network (WSN) and low-cost environmental sensors are thus used. In the specialized literature, several studies have been carried out to highlight the advantages and limitations of these technologies. In [4] the methodologies for designing a real-time air pollution

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monitoring system were presented, as well as the limits of existing technologies to achieve future air monitoring systems more accurate and realistic. Thus, a compromise between the quality of the measurements, the energy efficiency and the low cost of the monitoring system must be found.

Due to the size and the cost of traditional air quality monitoring stations, they are not ideal for dense deployment in urban areas [5]. Although accurate measurements of environmental parameters are possible, time-consuming offline processing is required. So, this method cannot provide real-time accurate air quality data. The rapid development of Internet-of-Things (IoT) technology enables the detection of air quality and the transmission of associated data to servers through wireless sensor networks (WSN) [6].

Many inexpensive portable air quality sensors are also used. For example, PM sensors are a very good option to provide accurate data on the amount of dust in the air in cities [7] within seconds. Although the accuracy of these sensors is not as high as that of the conventional air quality monitoring equipment, it is sufficient to demonstrate the trend of air quality change.

However, it is known that WSN technology has a short distance data transmission, it has a very limited coverage and therefore cannot support the communication needs of a large number of air quality sensors distributed over large areas. To overcome the problems mentioned above, LPWA (low power wide area) technology, a new type of Machine-to-Machine (M2M) communication, is used in [3]. It is designed to provide better coverage, low power consumption and cheap prices for devices that send and receive messages [8]. In general, an LPWA network can provide suburban and rural coverage of about 20 km and typical urban coverage of about 5 km. In addition, LPWA technology allows devices to operate for up to 10 years on a single charge.

These features of LPWA technology make it particularly suitable for typical smart city applications such as urban air quality monitoring.

In [9] the authors propose an energy-efficient IoT air quality monitoring system, describing the proposed hardware and software components. To check the air quality, the system was developed based on environmental sensors and a battery.

III. Prepare Your Paper Before Styling

In specialized literature, in international standards and in most environmental monitoring systems, an Air Quality Index (AQI) is used as a measure of air pollution to assess and manage the current level of air quality and public presence large. This index is calculated using formulas and concentrations that track at least three relevant air contaminants [9]. In general, air quality standards are updated, pollutants are added or modified from the existing list and examined periodically. Currently, the primary pollutants considered significant are: PM10 particles, PM2.5 particles, ozone (O3), Nitrogen dioxide (NO2), Sulfur dioxide (SO2) and carbon monoxide (CO).

The term "particulate matter" or PM for short, refers to the mixture of solid and liquid particles that can be found in either dry or moist air. PM is also a common synonym for "particle pollution". Due to their microscopic size, these particles are invisible to the naked eye. Most particulate matter (PM) is composed of nitrates, soot, rubber, metals, sulfates, water, dust, and other particles. PM2.5 and PM10 are terms used to describe particles of matter in the atmosphere that are 2.5 mm and 10 mm in diameter [9].

As in the specialized literature there are several methods of calculating AQI, for our study we used in the first phase a basic formula for calculating the air quality index [10], followed by the European index called CAQI (Common Air Quality Index) [11,12], and the quality index offered by the EPA [13]. Based on all these three formals AQI was computed and compared. Finally, a comparison with the European and American standards, as well with the limits imposed by the European Union and the WHO were discussed.

A. *Basic Air Quality Index - AQI*

The basic formula for calculating the air quality index is as follows:

$$AQI_i = \frac{C_i}{T_i} * 100 \tag{1}$$

where T_i is the maximum allowable limit per air pollutant standard, C_i is the monitored concentration of pollutant I measured over a 24h time interval, and AQI_i is the individual air quality index of pollutant i. The multiplication to 100 is utilized in equation (1) to improve the index value's intuitiveness.

The value representing the overall air quality index *(AQI)* can be acquired using [9]:

$$AQI = max(AQI_i) \tag{2}$$

The accepted values for AQI are set in Table I.

TABLE I. TABLE WITH POLLUTION THRESHOLD VALUES [9]

AQI	AQI		
	PM 2.5 μg/m³	***PM 10 μg /m³***	***Status***
0 – 50	0 – 30	0 – 50	Good
51 – 100	31 – 60	51 – 100	Satisfactory
101 – 200	61 – 90	101 – 250	Moderate
201 – 300	91 – 120	251 – 350	Low
301 – 400	121 – 250	351 – 430	Very Low
401 – 500	250+	430+	Critical

B. *EU Air Quality Index - CAQI*

The main aim of CAQI was to develop an index which, without replacing the indices used locally within the European Union, would encourage meaningful comparisons within the EU. CAQI is a number between 1 and 100. A low CAQI number denotes good air quality, while a high number denotes poor air quality. Hourly and daily descriptions of the index are used. Currently, the three required components of CAQI are NO2, PM10 and O3. Moreover, SO2, CO and PM2.5 were optional pollutants. A "subindex" is calculated for each of the required elements. CAQI represents the lowest quality sub-index of these elements. In this study, the time interval was used to measure each sub-index and to calculate the overall CAQI.

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C. EPA Air Quality Index

Another way of assessing air quality is supported by the Environmental Protection Agency (EPA), a federal government agency of the United States of America whose task is the application and development of environmental policy.

The main formula used by the EPA in calculating the AQI is as follows:

$$I = \frac{I_{High} - I_{Low}}{C_{High} - C_{Low}} (C - C_{Low}) - I_{Low} \quad (3)$$

In this formula I is the index, C is the pollutant concentration C_{Low} is the concentration breakpoint less than or equal equal to C, C_{High} is the concentration breakpoint greater than or equal to C, I_{Low} is the index breakpoint corresponding to C_{Low} , and I_{High} signifies the index breakpoint corresponding to C_{High}.

TABLE II. CAQI AIR QUALITY MONITORING STANDARD VALUES [9]

NO_2 µ g/m³	PM_{10} µg /m³	O_3 µ g/m³	$PM_{2.5}$ (optional) µg/m³	Index or Sub-index	Status
0–50	0–25	0 60	0–15	0–25	Very Low
50–100	25–50	60 120	15–30	25–50	Low
100–200	50–90	120 180	30–55	50–75	Moderate
200–400	90–180	180 240	55–110	75–100	High
>400	>180	>240	>110	>100	Very High

TABLE III. EPA TABLE OF BREAKPOINTS [14]

O_3 (ppb)	O_3 (ppb)	$PM_{2.5}$ (µg/m³)	PM_{10} (µg/m³)	CO (ppm)	SO_2 (ppb)	NO_2 (ppb)	AQI	Status
$C_{low} – C_{high}$ (avg)	$C_{low} – C_{high}$ (avg)	$C_{low} – C_{high}$ (avg)	$C_{low} – C_{high}$ (avg)	$C_{low} – C_{high}$ (avg)	$C_{low} – C_{high}$ (avg)	$C_{low} – C_{high}$ (avg)	$I_{low} – I_{high}$	Category
0–54 (8-hr)	—	0.0–12.0 (24-hr)	0–54 (24-hr)	0.0–4.4 (8-hr)	0–35 (1-hr)	0–53 (1-hr)	0–50	Good
55–70 (8-hr)	—	12.1–35.4 (24-hr)	55–154 (24-hr)	4.5–9.4 (8-hr)	36–75 (1-hr)	54–100 (1-hr)	51–100	Moderate
71–85 (8-hr)	125–164 (1-hr)	35.5–55.4 (24-hr)	155–254 (24-hr)	9.5–12.4 (8-hr)	76–185 (1-hr)	101–360 (1-hr)	101–150	Unhealthy for sensitive groups of people
86–105 (8-hr)	165–204 (1-hr)	55.5–150.4 (24-hr)	255–354 (24-hr)	12.5–15.4 (8-hr)	186–304 (1-hr)	361–649 (1-hr)	151–200	Unhealthy
106–200 (8-hr)	205–404 (1-hr)	150.5–250.4 (24-hr)	355–424 (24-hr)	15.5–30.4 (8-hr)	305–604 (24-hr)	650–1249 (1-hr)	201–300	Very Unhealthy
—	405–504 (1-hr)	250.5–350.4 (24-hr)	425–504 (24-hr)	30.5–40.4 (8-hr)	605–804 (24-hr)	1250–1649 (1-hr)	301–400	Hazard
—	505–604 (1-hr)	350.5–500.4 (24-hr)	505–604 (24-hr)	40.5–50.4 (8-hr)	805–1004 (24-hr)	1650–2049 (1-hr)	401–500	

D. Comparison EU and WHO Standards

European Union (EU) directives on air quality, in particular Directive 2008/50/EC on ambient air quality and cleaner air for Europe and Directive 2004/107/EC on heavy metals and polycyclic aromatic hydrocarbons in ambient air, set limits for pollutant concentrations and for concentrations that cannot be exceeded within a certain time interval. The following figure summarizes some EU and World Health Organization (WHO) norms. These criteria and limits are applicable over different periods of time because the health effects of different contaminants manifest themselves over different periods of time.

As the WHO guidelines are created with the intention of protecting public health, they are stricter than the corresponding EU norms, which are more widely influenced by the political environment [18].

		EU Air Quality Directives			WHO Air Quality Guidelines					
Pollutant	Averaging period	Objective	Concentration	Comments	Concentration					Comments
					Interim targets				AQG level	
					1.	2.	3.	4.		
$PM_{2.5}$	24-hour	Target value			75	50	37,5	25	15 µg/m³	99th percentile (i.e. 3–4 exc. Days/year)
$PM_{2.5}$	Annual	Limit value	25 µg/m³		35	25	15	10	5 µg/m³	
$PM_{2.5}$	Annual	Indicative limit value	20 µg/m³							
PM_{10}	24-hour	Limit value	50 µg/m³	Not to be exceeded on more than 35 days/year	150	100	75	50	45 µg/m³	99th percentile (i.e. 3–4 exc. Days/year)
PM_{10}	Annual	Limit value	40 µg/m³		70	50	30	20	15 µg/m³	
O_3	Max. daily 8-hour mean	Target value	120 µg/m³	Not to be exceeded on more than 25 days/year (averaged over 3 years)						
O_3	Max. daily 8-hour mean	Long-term objective	120 µg/m3							
O_3	8-hour	Target value			160	120	–	–	100 µg/m³	99th percentile (i.e. 3–4 exc. Days/year)
O_3	Peak season [3]	Target value			100	70	–	–	60 µg/m³	
NO_2	Hourly	Limit value	200 µg/m³	Not to be exceeded on more than 18 hours/year					200 µg/m³	
NO_2	Annual	Limit value	40 µg/m³		40	30	20	–	10 µg/m³	
NO_2	24-hour	Target value			120	50	–	–	25 µg/m³	99th percentile (i.e. 3–4 exc. Days/year)
SO_2	Hourly	Limit value	350 µg/m³	Not to be exceeded on more than 24 hours/year						
SO_2	24-hour	Limit value	125 µg/m³	Not to be exceeded on more than 3 days/year	125	50	–	–	40 µg/m³	99th percentile (i.e. 3–4 exc. Days/year)
CO	Max. daily 8-hour mean	Limit value	10 mg/m³						10 mg/m³	
CO	24-hour	Target value			7	–	–	–	4 mg/m³	99th percentile (i.e. 3–4 exc. Days/year)
C_6H_6	Annual	Limit value	5 µg/m³						1,7 µg/m³	Reference level
BaP	Annual	Target value	1 ng/m³	Measured as content in PM_{10}						
Pb	Annual	Limit value	0,5 µg/m³	Measured as content in PM_{10}					0,5 µg/m³	
As	Annual	Target value	6 ng/m³	Measured as content in PM_{10}					6,6 ng/m³	Reference level
Cd	Annual	Target value	5 ng/m³	Measured as content in PM_{10}					5 ng/m³	
Ni	Annual	Target value	20 ng/m³	Measured as content in PM_{10}					25 ng/m³	Reference level

Fig. 1. EU and WHO Standards for AQI [15,16]

IV. PROPOSED APPLICATION FOR AQI MONITORING

Considering the different ways of calculating air quality indices, the different ways of acquiring, processing and interpreting data, an application was further developed as a modern, eloquent and useful tool in solving the problems of monitoring environmental parameters. The application is based on Power Platform tools such as PowerApps and Power Automate Flow. Some benefits for this app are:

- Extremely low cost.
- Adaptability, flexibility and portability.
- Connection with many equipment and databases.
- The possibility to create a customized communication gateway.
- Increased data security and the possibility to work with big data.
- The easy to be shared with other members of the organization, and more.

This application uses 3 open-source data sets [17,18,19], which we can access through APIs that are available on the websites of the respective sources. These three sources aim to provide data about environmental parameters in real time, and through custom databases and gateways made with Power Automate Flow, that data is retrieved and displayed in real time in the developed application. The real time environment

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parameters are collected from the open-source databases and are archived in cloud to create a history and compare this data. It should be noted that all this captured data is used strictly for research and analysis purposes, not for any other purpose. Furthermore, this historical data are used for AQI calculation by our software application using different standards.

The developed application has a friendly and intuitive interface, with the help of which users can perform different actions to monitor the environmental parameters and assign meaning to these data. Users can check real-time environmental data for a region of interest (in this study we will analyze all data for the city of Bucharest, Romania). They can see a specific history using a start and end date (for this study we will consider a period of 24 hours for each individual source, but the app has the ability to display a longer history, up to months/years if needed).

Through the application, the data are taken from the three mentioned sources, displayed also in the form of graphs for a better interpretation of them, and based on the calculation methods presented, it is possible to identify the AQI and the degree of risk to which we are subjected if we breathe a lot of time the air in that area according to the standards presented. Practically in a single application we can put together several data sources, we will check the accuracy of the indices established by the sources by means of the implemented calculation formulas, then the results obtained are compared with the additive limits established by different standards to assess the degree of risk we are at subjects, but also to validate those sources. As a brief overview of what follows in this chapter, it should be mentioned that the data that we can see in figures 2 and 3(a,b,c) are strictly data taken from our sources, without any additional processing. The data collected from source 1 are used to provide information for air quality monitoring in figure 2.

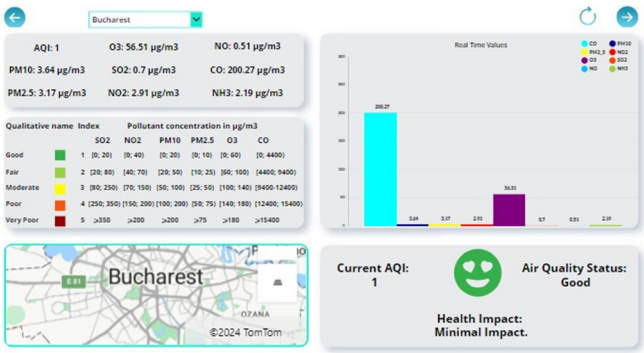

Fig. 2. Interface with all data from one of the sources in real time

It should be mentioned that the first data source numbers the AQI based on 5 steps from 1 to 5, the thresholds being chosen according to the values represented in the table in figure 2.

Contrary to this, the other 2 sources do not use a general step, but a number that is the result based on the maximum of the values of the pollutants of interest. Here we can observe the first important difference between the respective sources, namely the way of recording and interpreting the data, which will be seen in one of the future figures where we can compare all these data.

Considering for all 3 open data sources the same day, 2nd April the difference between considered measured pollutants and AQI calculated are varying. In the following pictures PM10 is considered daily with information at each hour, but similar data analysis could be done for all monitored pollutants (PM2 .5, SO2, NO2, CO, O3). The level of pollution fluctuated during a day, at the same time to see the differences in values between the 3 sources (Figures 3, 4, 5).

Using the AQI from the three open data sources, for the selected 24 hours, and establish an average AQI for that day, as well as its minimum and maximum the next figure (Figure 6 and 7) allow as to see the difference in AQI value and interpretation accordingly to the limits establish for qualitative name index tables in figure 2.

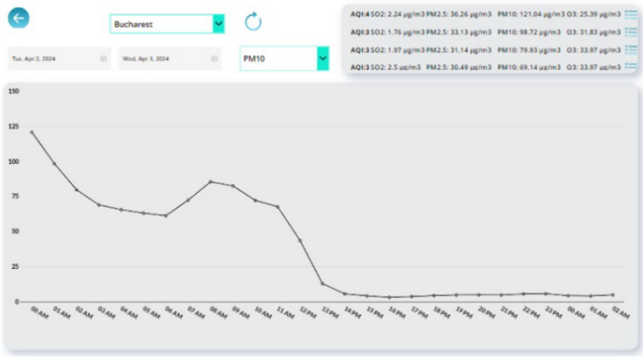

Fig. 3. Data collected from source 1 on 2nd April and representation for PM10 in that day

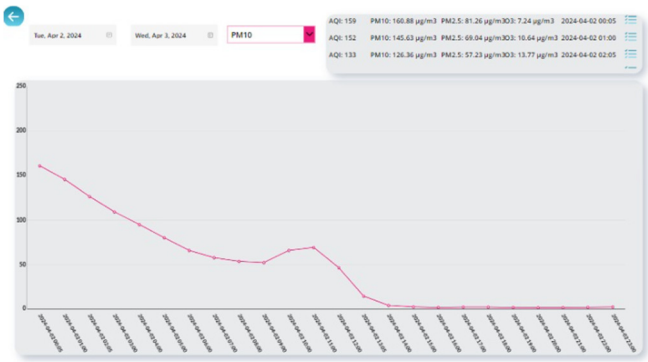

Fig. 4. Data collected from source 2 on 2nd April and representation for PM10 in that day

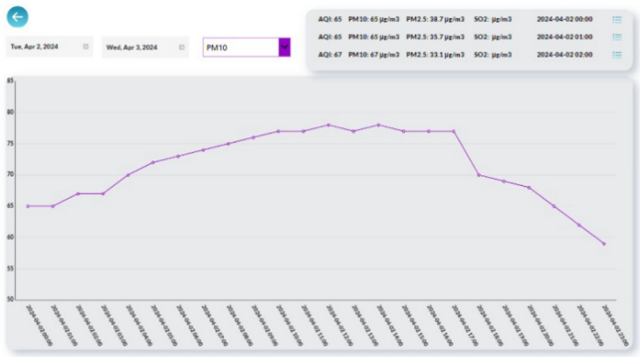

Fig. 5. Data collected from source 3 on 2nd April and representation for PM10 in that day

To obtained un unified representation of the AQI index for all considered data sources, the AQI index is computed with the first basic method (eq.1 and Table 1). For data source 1 the results are represented in figure 8.

Starting with figure 6, the obtained results are computed after processing the data. To explore more on the significance

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of AQI significance the second method was implemented on the 3 available data sources. The result of CAQI calculus is using eq 3. and Table 2. The next figure 7 present the results and the comparison with allowed standard limits.

Comparing the result in figure 8 and 9 the significance of AQI is slightly different, the first method of calculating the AQI shows that it is somewhere between good and moderate, but in the second method of calculation (CAQI) it is observed that the result is mainly between medium and high.

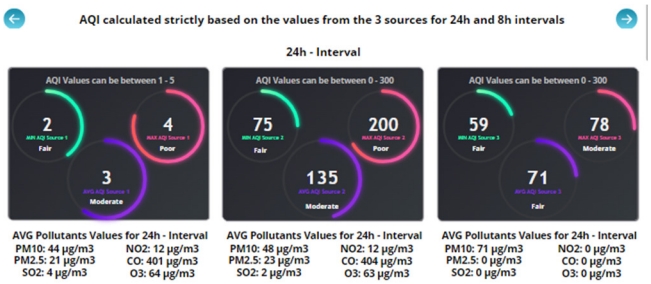


Fig. 6. Values obtained from data without additional processing on a 24h interval

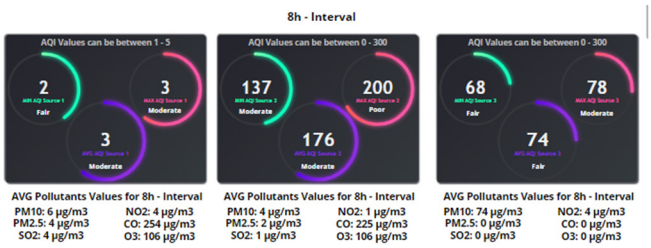


Fig. 7. Values obtained from data without additional processing on a 8h interval

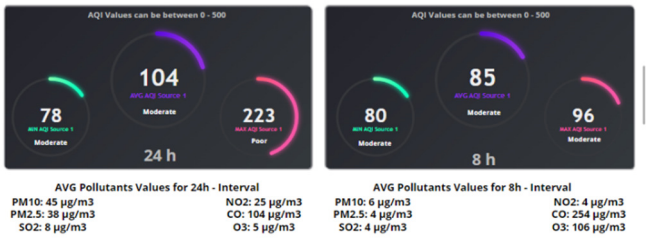
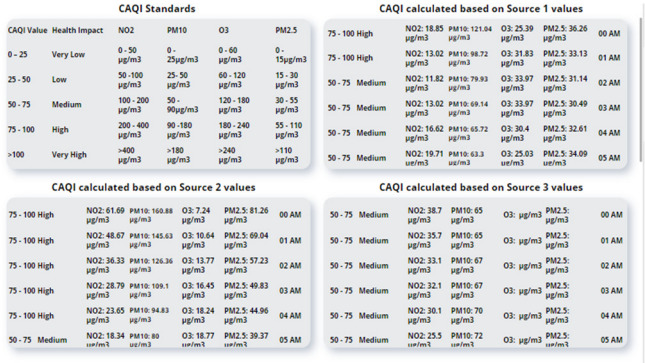


Fig. 8. Result obtained for source one after implementation of the AQI basic formula

Finally, the AQI index is calculated with the 3rd formula, the one offered by the EPA. Again, the same conclusion based on all open-source data is find, namely that the air in our area of interest is Very unhealthy according to the data 1 and 2 and Moderate according to date 3 (figure 10), which is otherwise true and easy to intuit considering the fact that Bucharest is one of the most crowded cities from Romania with different sources of atmospheric pollution as studied in [20].

The conclusion of this comparison is that although a lot of effort and resources of all kinds are invested to reduce pollution, it can still be clearly observed that all 3 sources indicate that the pollution values are still above the limits accepted by the two organizations.

As we said before, the next step is to compare the values previously obtained from the data sources with the EU and WHO standards, the obtained results can be seen in the following figure 11.



CAQI Standards

CAQI Value	Health Impact	NO2	PM10	O3	PM2.5
0 - 25	Very Low	0 - 50 µg/m3	0 - 25µg/m3	0 - 60 µg/m3	0 - 15µg/m3
25 - 50	Low	50 -100 µg/m3	25- 50 µg/m3	60 - 120 µg/m3	15 - 30 µg/m3
50 - 75	Medium	100 - 200 µg/m3	50 - 90µg/m3	120 - 180 µg/m3	30 - 55 µg/m3
75 - 100	High	200 - 400 µg/m3	90 -180 µg/m3	180 - 240 µg/m3	55 - 110 µg/m3
>100	Very High	>400 µg/m3	>180 µg/m3	>240 µg/m3	>110 µg/m3

CAQI calculated based on Source 1 values

75 - 100 High	NO2: 18.85 µg/m3	PM10: 121.04 µg/m3	O3: 25.39 µg/m3	PM2.5: 36.26 µg/m3	00 AM
75 - 100 High	NO2: 13.02 µg/m3	PM10: 98.72 µg/m3	O3: 31.83 µg/m3	PM2.5: 33.13 µg/m3	01 AM
50 - 75 Medium	NO2: 11.82 µg/m3	PM10: 79.93 µg/m3	O3: 33.97 µg/m3	PM2.5: 31.14 µg/m3	02 AM
50 - 75 Medium	NO2: 13.02 µg/m3	PM10: 69.14 µg/m3	O3: 33.97 µg/m3	PM2.5: 30.49 µg/m3	03 AM
50 - 75 Medium	NO2: 16.62 µg/m3	PM10: 65.72 µg/m3	O3: 30.4 µg/m3	PM2.5: 32.61 µg/m3	04 AM
50 - 75 Medium	NO2: 19.71 ug/m3	PM10: 63.3 ug/m3	O3: 25.03 ug/m3	PM2.5: 34.09 ug/m3	05 AM

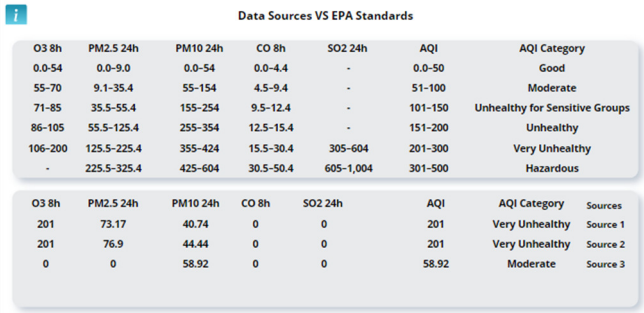
CAQI calculated based on Source 2 values

75 - 100 High	NO2: 61.69 µg/m3	PM10: 160.88 µg/m3	O3: 7.24 µg/m3	PM2.5: 81.26 µg/m3	00 AM
75 - 100 High	NO2: 48.67 µg/m3	PM10: 145.63 µg/m3	O3: 10.64 µg/m3	PM2.5: 69.04 µg/m3	01 AM
75 - 100 High	NO2: 36.33 µg/m3	PM10: 126.36 µg/m3	O3: 13.77 µg/m3	PM2.5: 57.23 µg/m3	02 AM
75 - 100 High	NO2: 28.79 µg/m3	PM10: 109.1 µg/m3	O3: 16.45 µg/m3	PM2.5: 49.83 µg/m3	03 AM
75 - 100 High	NO2: 23.65 µg/m3	PM10: 94.83 µg/m3	O3: 18.24 µg/m3	PM2.5: 44.96 µg/m3	04 AM
50 - 75 Medium	NO2: 18.34 ug/m3	PM10: 80 ug/m3	O3: 18.77 ug/m3	PM2.5: 39.37 ug/m3	05 AM

CAQI calculated based on Source 3 values

50 - 75 Medium	NO2: 38.7 µg/m3	PM10: 65 µg/m3	O3: µg/m3	PM2.5: µg/m3	00 AM
50 - 75 Medium	NO2: 35.7 µg/m3	PM10: 65 µg/m3	O3: µg/m3	PM2.5: µg/m3	01 AM
50 - 75 Medium	NO2: 33.1 µg/m3	PM10: 67 µg/m3	O3: µg/m3	PM2.5: µg/m3	02 AM
50 - 75 Medium	NO2: 32.1 µg/m3	PM10: 67 µg/m3	O3: µg/m3	PM2.5: µg/m3	03 AM
50 - 75 Medium	NO2: 30.1 µg/m3	PM10: 70 µg/m3	O3: µg/m3	PM2.5: µg/m3	04 AM
50 - 75 Medium	NO2: 25.5 ug/m3	PM10: 72 ug/m3	O3: µg/m3	PM2.5: ug/m3	05 AM

Fig. 9. Result obtained for source one after implementation of the AQI basic formula

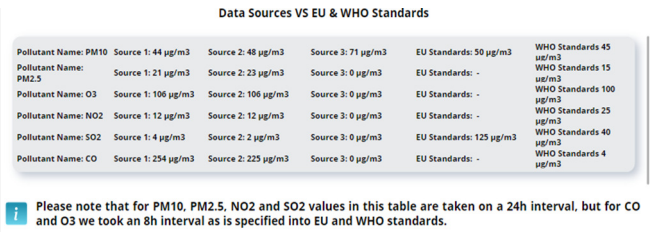


Data Sources VS EPA Standards

O3 8h	PM2.5 24h	PM10 24h	CO 8h	SO2 24h	AQI	AQI Category
0.0-54	0.0–9.0	0.0–54	0.0–4.4	-	0.0–50	Good
55–70	9.1–35.4	55–154	4.5–9.4	-	51–100	Moderate
71–85	35.5–55.4	155–254	9.5–12.4	-	101–150	Unhealthy for Sensitive Groups
86–105	55.5–125.4	255–354	12.5–15.4	-	151–200	Unhealthy
106–200	125.5–225.4	355–424	15.5–30.4	305–604	201–300	Very Unhealthy
-	225.5–325.4	425–604	30.5–50.4	605–1,004	301–500	Hazardous

O3 8h	PM2.5 24h	PM10 24h	CO 8h	SO2 24h	AQI	AQI Category	Sources
201	73.17	40.74	0	0	201	Very Unhealthy	Source 1
201	76.9	44.44	0	0	201	Very Unhealthy	Source 2
0	0	58.92	0	0	58.92	Moderate	Source 3

Fig. 10. Result obtained for source one after implementation of the AQI basic formula



Data Sources VS EU & WHO Standards

Pollutant Name: PM10	Source 1: 44 µg/m3	Source 2: 48 µg/m3	Source 3: 71 µg/m3	EU Standards: 50 µg/m3	WHO Standards 45 µg/m3
Pollutant Name: PM2.5	Source 1: 21 µg/m3	Source 2: 23 µg/m3	Source 3: 0 µg/m3	EU Standards: -	WHO Standards 15 µg/m3
Pollutant Name: O3	Source 1: 106 µg/m3	Source 2: 106 µg/m3	Source 3: 0 µg/m3	EU Standards: -	WHO Standards 100 µg/m3
Pollutant Name: NO2	Source 1: 12 µg/m3	Source 2: 12 µg/m3	Source 3: 0 µg/m3	EU Standards: -	WHO Standards 25 µg/m3
Pollutant Name: SO2	Source 1: 4 µg/m3	Source 2: 2 µg/m3	Source 3: 0 µg/m3	EU Standards: 125 µg/m3	WHO Standards 40 µg/m3
Pollutant Name: CO	Source 1: 254 µg/m3	Source 2: 225 µg/m3	Source 3: 0 µg/m3	EU Standards: -	WHO Standards 4 µg/m3

Please note that for PM10, PM2.5, NO2 and SO2 values in this table are taken on a 24h interval, but for CO and O3 we took an 8h interval as is specified into EU and WHO standards.

Fig. 11. Result obtained for source one after implementation of the AQI basic formula

V. Conclusion

This work aimed to present a synthesis for outdoor air quality monitoring system, considering new technologies and facilities. The main contribution of this paper was the developed concept of a low-cost air quality monitoring system based on low-cost sensors for environment parameters considered by international standards, IoT technology and the software application.

Considering the types of monitored pollutants, the methods of calculating the air quality index (using standards and available data sources), the healthy admissible thresholds, and the available data a software application based on modern technologies was designed, implemented, and validated. Considering the multitude of the proposed app functionalities and complexity, no similar software application was found. This instrument offers a versatile tool for validating and interpreting the data that a citizen can use from public sources. He can validate the information according to the standards in force and compare the information provided by different sources between them.

The obtained results show that the different available public sources do not have a clear calculation formula that

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follows a certain standard, there are differences between the quality indices suggested by the available sources and the quality indices calculated by the formulas presented on the same data sets. Moreover, the 3 sets have differences in standard AQI results, some of them being even significant (we can observe the differences between the first 2 sources and the 3rd one where the value of the index for source 3 is almost 2 times lower than in the other 2 cases in Fig. 8). The AQI is represented with a significant difference, one source prefers steps from 1 to 5, while the other 2 sources interpolate between 0 - 500, leading to inconsistency and perhaps even a degree of confusion for users. However, there are also similar aspects for the considered sources, more precisely, processing the data we notice that regardless of the calculation model used, the result suggests AQI values above the allowed value, which is a factor of concern at this moment.

This work is part of a longer research period, in which we are already laying the foundations of a system for monitoring environmental parameters that aims to be low-cost and energy independent. In further developments the system will provide a series of data from its own sensors, will calculate AQI according to various standards and will be able to make comparisons with the available sources.

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